Yibo Yuan

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Research interests: computational and systems neuroscience; naturalistic social behavior; neural population dynamics; closed-loop neuroengineering

EDUCATION

Xi'an Jiaotong University, Xi'an, China - Sep 2021 - Jun 2025

B.Eng. in Biomedical Engineering, School of Life Science and Technology

Selected coursework: Pattern Recognition and Machine Learning (85/100); Biomedical Sensing and Instrumentation (84/100); Data Structures and Algorithms (80/100); Medical Device R&D Management and Product Certification (100/100)

RESEARCH APPOINTMENTS

Westlake University, Social Neuroscience Network Laboratory - Aug 2025 - Present

Full-time Research Assistant, Prof. Ding Liu - Hangzhou, China

Illinois Institute of Technology, TIML Research Lab - 2025 - Present

Part-time Research Intern, Prof. Ren Wang - Remote

PUBLICATION

Dai, Y. Y., Wei, C., **Yuan, Y.**, Rahman, M. M., & Liu, D. (2026). In vivo calcium imaging with a miniaturized microscope in the hypothalamus for understanding social behaviors in mice. *Journal of Visualized Experiments*, (229). doi:10.3791/70401

SELECTED RESEARCH EXPERIENCE

Computational Analysis of Naturalistic Social Behavior, Westlake University - Aug 2025 - Present

- Independently developed an open-source behavioral-video analysis pipeline supporting irregular multi-region tracking, temporal alignment, perspective correction, frame-level outputs, and semantic-segmentation-based contact detection.
- Developed hazard-function analyses of social-contact duration and multimetric analyses of neural population geometry, including discriminability, statistical divergence, subspace alignment, gain modulation, CKA, and time-shift permutation controls.
- Current analyses test how social isolation and reunion alter hypothalamic population representations during free social interaction.
- Trained in miniscope deployment, stereotactic injection, optogenetic manipulation, mouse perfusion, and brain cryosectioning.

Experimental Platforms for Neural and Behavioral Recording, Westlake University - Aug 2025 - Present

- Built an open-source 8-channel STM32 lickometer with UART transmission and a Python host interface for event visualization and CSV storage.
- Designed and assembled a programmable three-axis motion platform with stepper-motor control and configurable motion profiles for motion-sickness experiments.
- Designed and 3D-printed experimental components and integrated behavioral monitoring with miniscope and physiological recording workflows.

Cross-Subject Multimodal EEG Modeling, Illinois Institute of Technology - 2025 - Present

- Reviewed multimodal EEG datasets and developed preprocessing and cross-subject baseline workflows for affective-state classification.
- Evaluating episodic meta-learning and early, late, and contrastive fusion strategies for subject adaptation under limited calibration data.

SELECTED RESEARCH EXPERIENCE - CONTINUED

Programmable Social-Stimulus Platform, Beijing Institute of Technology collaboration - Jan 2026 - Present

- Contributed a control-theoretic formulation of social homeostasis and designed a programmable embodied stimulus with tunable motion, tactile, thermal, and olfactory cues.
- Developing the platform and closed-loop vision-control workflow; animal and neural validation remain future work.

Portable Bioimpedance Spectroscopy System, Xi'an Jiaotong University - Jan 2025 - Jun 2025

- Designed a tissue-water-content sensing prototype based on multi-frequency bioimpedance spectroscopy and the Cole-Cole model.
- Integrated STM32 control, an AD5933 impedance-converter front end, a two-electrode acquisition circuit, PCB design, hardware debugging, and benchtop testing.

Eight-Channel EEG Acquisition System, Xi'an Jiaotong University - Mar 2024 - Sep 2024

- Led PCB design for an ADS1299/STM32F103 acquisition system, including power/reference circuits, input protection, anti-aliasing, bias drive, and test-signal paths.
- Completed power, reference, and clock checks and specified the remaining SPI, DRDY, shorted-input noise, CMRR, and signal-validation plan.

OPEN-SOURCE ENGINEERING

- **NeuronTracker:** OpenCV/Tkinter tool for semi-automated axon-growth tracking with waypoint, direction, smoothness, and step-size constraints; exports trajectories and calibrated growth-speed tables.
- **Mouse-trajectory-tracking:** Behavioral-video analysis for custom ROIs, long-duration tracking, quantitative CSV summaries, and experimental multi-animal contact detection.
- **Lickometer:** STM32 firmware and Python acquisition interface for eight-channel capacitive lick detection.

SELECTED HONORS

- National-level College Students' Innovation and Entrepreneurship Training Program (2024 and 2023).
- Second Prize, 35th Tengfei Cup Innovation and Entrepreneurship Competition (2024); Second Prize, Xi'an Jiaotong University Mathematical Modeling Competition (2023).
- Gold Medal, International Genetically Engineered Machine Competition (iGEM), 2022.

TECHNICAL SKILLS

Programming and analysis: Python, C, MATLAB; PyTorch, OpenCV, MMSegmentation, scikit-learn, MNE, pandas, NumPy
Hardware: STM32, ADS1299, AD5933, UART, PCB design, soldering/debugging, stepper-motor control, Blender/3D printing
Neuroscience: behavioral video analysis, miniscope workflows, stereotactic surgery, perfusion, cryosectioning, optogenetic manipulation